INCH-POUND

MIL-DTL-25666/3 (USAF)
15 September 1995
SUPERSEDING
MIL-C-25974D (USAF)
01 April 1983

DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-10/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1 Scope

1.1 Scope. This specification covers one type of liquid oxygen converter with a capacitance gauging element. The converter provides storage and pressurization of 10 liters of liquid oxygen to 300 pounds per square inch gauge (psig), nominal pressure and does not include warming coils. The converter is designated type GCU-10/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-DTL-25666	Converter, Liquid Oxygen, Capacitance Type Gauging, General Specification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-5990 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129	Marking for Shipment and Storage
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Unless otherwise indicated, copies of the above specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Qualification. The converters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design. The converter shall conform to the dimensions and schematic diagram specified in figures 1, 2, 3, and 5. Port and electrical connector locations and angles shall be as specified in table II. Reference planes in table II are defined in figure 4. The converter shall have a minimum capacity of 10 liters of liquid oxygen measured at 1 atmosphere, plus at least 3% of this volume to allow for the change in liquid density at the operating pressure. The converter shall be designed for an operating pressure of 300 psig.

3.2.1 Design strength. The converter when full shall be designed to withstand the following g forces without damage:

Vertical (down)	14G
Vertical (up)	6G
Lateral	6G

3.2.2 Relief valve. The relief valves shall conform to the performance requirements of types III and IV of MIL-V-9050 and shall be of such size and shape to fit the over all dimensions specified for the converter.

3.2.3 Weight. The weight of the complete converter shall not exceed 18 pounds.

3.2.4 Marking of ports. Ports shall be marked in accordance with figure 1.

3.3 Performance.

3.3.1 Final assembly pressure. The pressure used for final assembly pressure shall be 320 psig.

3.3.2 Filling time. Filling time shall not exceed 10 minutes when tested as specified in MIL-DTL-25666.

3.3.3 Evaporation loss.

3.3.3.1 Vent condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed 3 pounds of oxygen when tested as specified in MIL-DTL-25666.

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3.3.3.2 Buildup condition. The evaporation loss of the essentially full converter in the buildup condition shall not exceed 3 pounds of oxygen when tested as specified in MIL-DTL-25666.

3.3.4 Buildup time. Buildup time shall not exceed 10 minutes as specified.

3.3.5 Capacitance. The capacitance of the sensing element of the empty converter shall be 123.5 ±1.0 picofarad (pf) when tested as specified in MIL-DTL-25666.

3.3.6 Inner vessel pressure.

3.3.6.1 Inner vessel. The inner vessel shall meet the performance requirements for the inner vessel pressure test specified in MIL-DTL-25666 when subjected to a pressure of 500 psig.

3.3.6.2 Permanent volumetric expansion. The converter shall meet the performance requirements for permanent volumetric expansion as specified in MIL-DTL-25666 when subjected to a pneumatic or hydrostatic pressure of 500 psig .

3.3.7 Delivery rate.

3.3.7.1 Constant flow . The converter shall deliver a flow of gaseous oxygen at a flow rate up to 80 liters per minute for not less than 1 hour with the operating pressure (measured in the gaseous phase of the system) remaining between 295 and 315 psig after the first 15 minutes of flow. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2 minute interval for any one pressure cycle. The flow at the specified rate shall be initiated within 10 minutes after the converter has reached the operating pressure.

3.3.8 Calibration. The capacitance sensing element shall have characteristics which provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 6, using the equations that are specified (see 3.4.6.1) to determine tolerances. An example of the use of these equations are as follows: If the actual quantity of liquid oxygen at the 6 liter test point is 6.1 liters, the measured capacitance must be between C maximum (max) and C minimum (min).

C min = 5.7 x 6.1 + 122.3 = 157.3 pf
C max = 5.9 x 6.1 + 124.7 = 160.7 pf

3.3.8.1 Equations. The following equations shall be used to calculate the maximum and minimum capacitances:

Values less than 2 liters	Values greater than 2 liters
C min = 5.6L + 122.5	C min = 5.7L + 122.3
C max = 6.0L + 124.5	C max = 5.9L + 124.7

C = Capacitance in PF
L = Exact quantity in liters of liquid at test point

3.3.9 Operating temperature. The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and between 290 and 315 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2 minute interval for any one pressure cycle.

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3.3.10 Vibration.

3.3.10.1 Vibration. The converter shall not exhibit physical evidence of damage after completion of the vibration tests as specified in MIL-DTL-25666.

3.3.10.2 Vibration - resonance. The converter shall not exhibit physical evidence of damage after completion of the resonance portion of the vibration test specified in MIL-DTL-25666.

3.3.10.3 Vibration - cycling. The converter shall not exhibit physical evidence of damage after completion of the cycling portion of the vibration test specified (see 4.4.1).

4. VERIFICATION

4.1 Classification of inspection. The examination and testing of the converter shall be classified as follows:

a. Qualification inspection (see 4.2)

c. Conformance inspection (see 4.3)

4.2 Qualification inspection. The qualification inspection shall consist of all the inspection methods specified in MIL-DTL-25666.

4.3 Conformance inspection. The conformance inspection shall consist of the individual tests and the sampling plans and tests specified in MIL-DTL-25666.

4.4 Vibration.

4.4.1 Vibration - cycling. The converter shall be vibrated under the cycling conditions specified herein for the applicable periods listed in table I. The frequency shall be cycled between 5 to 500 cps and return in a 15 minute cycle at an applied double amplitude of 0.036 inch or an applied acceleration of ±4.5G which ever is the lower G value. The rate of change of frequency shall be logarithmic (see 3.3.10.3).

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in the contract or order (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 10 liter, 300 psig converter covered by this specification is intended for use on F-104B aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25666 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from, OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-5990.

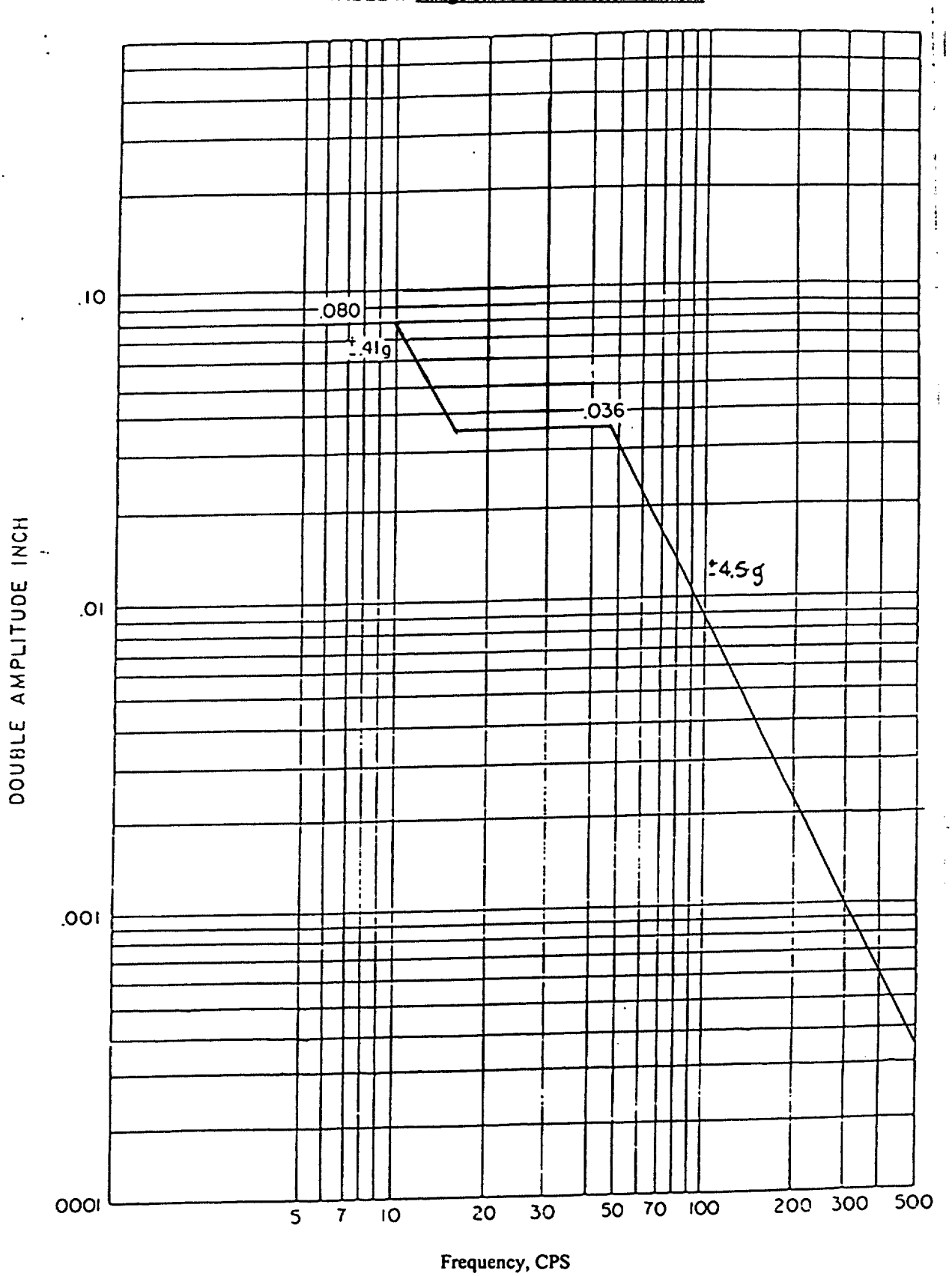
6.4 Subject term (key word) listing.

Breathing oxygen
Capacitance
Container, inner
Container, outer
Delivery rate
Dewar
Element, sensing
Evaporation loss
Gunfire
Life support
LOX
Valve, fill-buildup-vent
Valve, pressure closing
Valve, pressure relief
Vibration

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Range curve for converter vibration



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TABLE II. Port and electrical connector locations and angles.

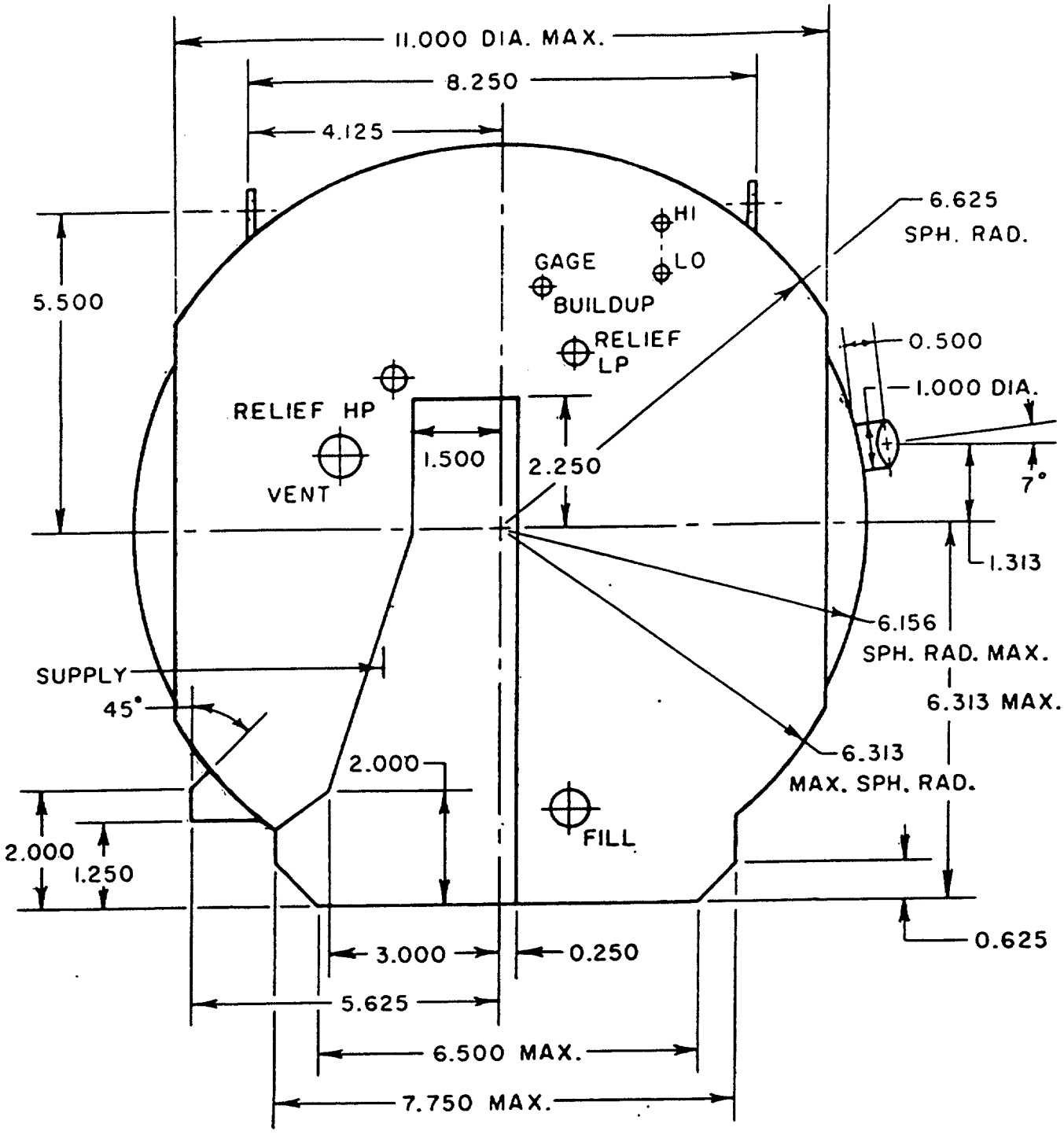
Ports	Size and type of threads	Location of center point of opening	Direction of boss centerline as projection on reference plane
Fill port	1/4 ANPT internal	1. 1 1/8 inches right of Reference Plane A 2. 5 7/16 inches in front of Reference Plane B 3. 4 3/4 inches below Reference Plane C	1. 40 degrees downward and forward in Reference Plane A 2. 13 degrees downward toward right in Reference Plane B 3. 13 degrees forward toward right in Reference Plane C
Supply port	MS33656-5	1. 1.915 inches left of Reference Plane A 2. 5 3/4 inches in front of Reference Plane B 3. 3 1/2 inches below Reference Plane C	1. A point in Reference Plane A 2. Horizontal line in Reference Plane B 3. Horizontal line in Reference Plane C
Build-up port	1/8 ANPT internal	1. 1/2 inch right of Reference Plane A 2. 6 9/16 inches in front of Reference Plane B 3. 4 11/32 inches above Reference Plane C	1. 25 degrees upward and forward in Reference Plane A 2. 45 degrees upward toward right in Reference Plane B 3. 60 degrees forward toward right in Reference Plan C
Vent port	3/8 ANPT internal	1. 2 11/16 inches left of Reference Plane A 2. 6 1/32 inches in front of Reference Plane B 3. 1 1/4 inches above Reference Plane C	1. 10 degrees upward and forward in Reference Plane A 2. 10 degrees upward toward left in Reference Plane B 3. 25 degrees forward toward left in Reference Plane C
Low pressure relief port	1/8 ANPT internal	1. 1 1/16 inches right of Reference Plane A 2. 6 inches in front of Reference Plane B 3. 3 3/16 inches above Reference Plane C	1. 30 degrees downward and forward in Reference Plane A 2. Vertical line in Reference Plane B 3. 90 degrees forward and back in Reference Plane C
High pressure relief port	1/8 ANPT internal	1. 2 5/32 inches left of Reference Plane A 2. 5 3/4 inches in front of Reference Plane B 3. 3 inches above Reference Plane C	1. 35 degrees downward and forward in Reference Plane A 2. Vertical line in Reference Plane B 3. 90 degrees forward and back in Reference Plane C

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TABLE II. Port and electrical connector locations and angles -Continued.

High impedance gauge connector	MIL-C-25516, B Polarity	1. 2 5/8 inches right of Reference Plan A 2. 4 31/32 inches in front of Reference Plane B 3. 5 1/16 inches above Reference Plane C	1. Horizontal line in Reference Plane A 2. Horizontal line in Reference Plane B 3. 30 degrees forward toward right in Reference Plane C
Low impedance gauge connector	MIL-C-25516, E Polarity	1. 2 5/8 inches right of Reference Plane A 2. 4 31/32 inches in front of Reference Plane B 3. 4 11/32 inches above Reference Plane C	1. Horizontal line in Reference Plane A 2. Horizontal line in Reference Plane B 3. 30 degrees forward toward right in Reference Plane C

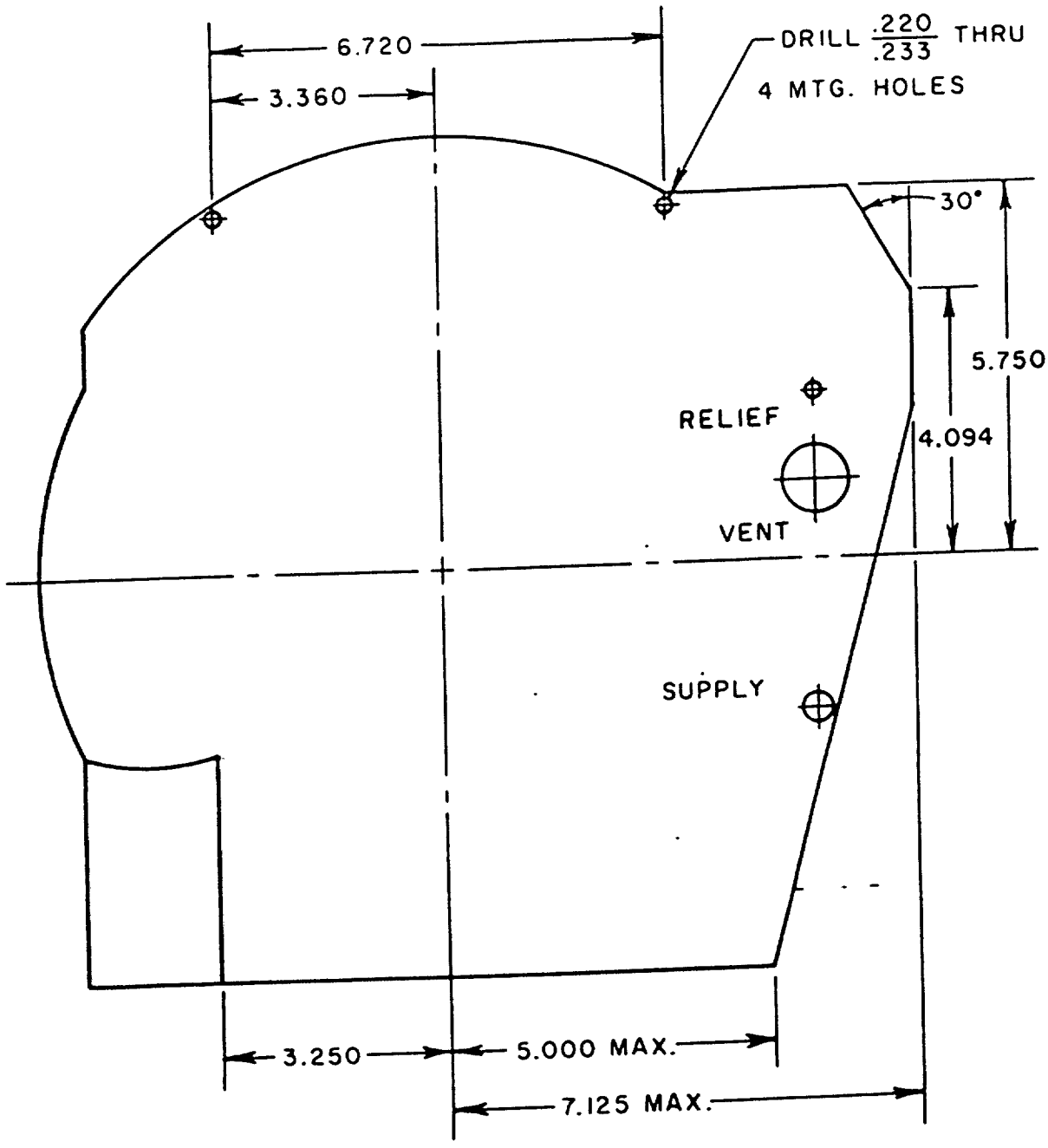
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UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCES ARE: DECIMALS ± 0.031 AND ANGLES ± 3°

FIGURE 1. Front view

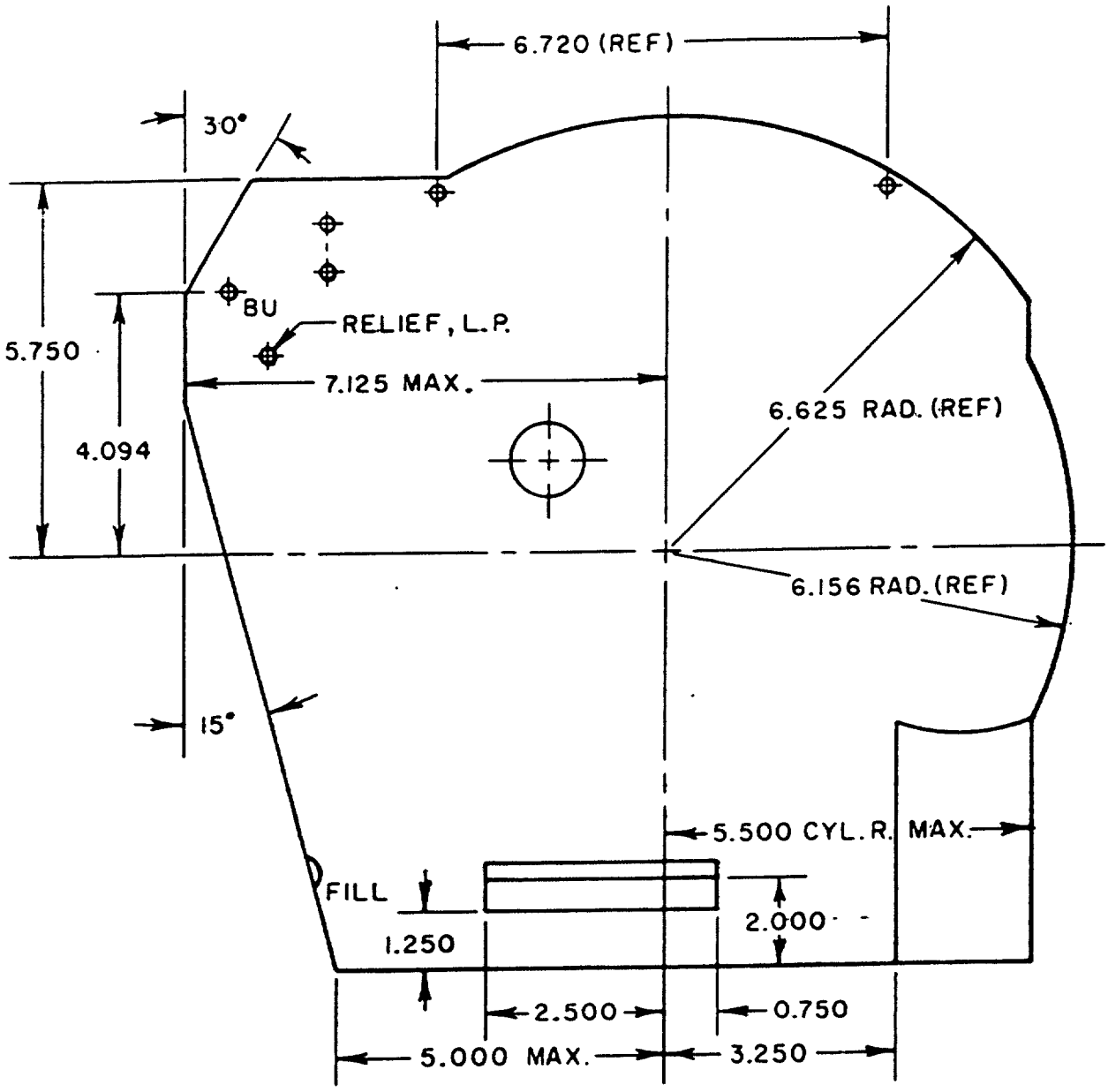
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FIGURE 2. Left side

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FIGURE 3. Right view

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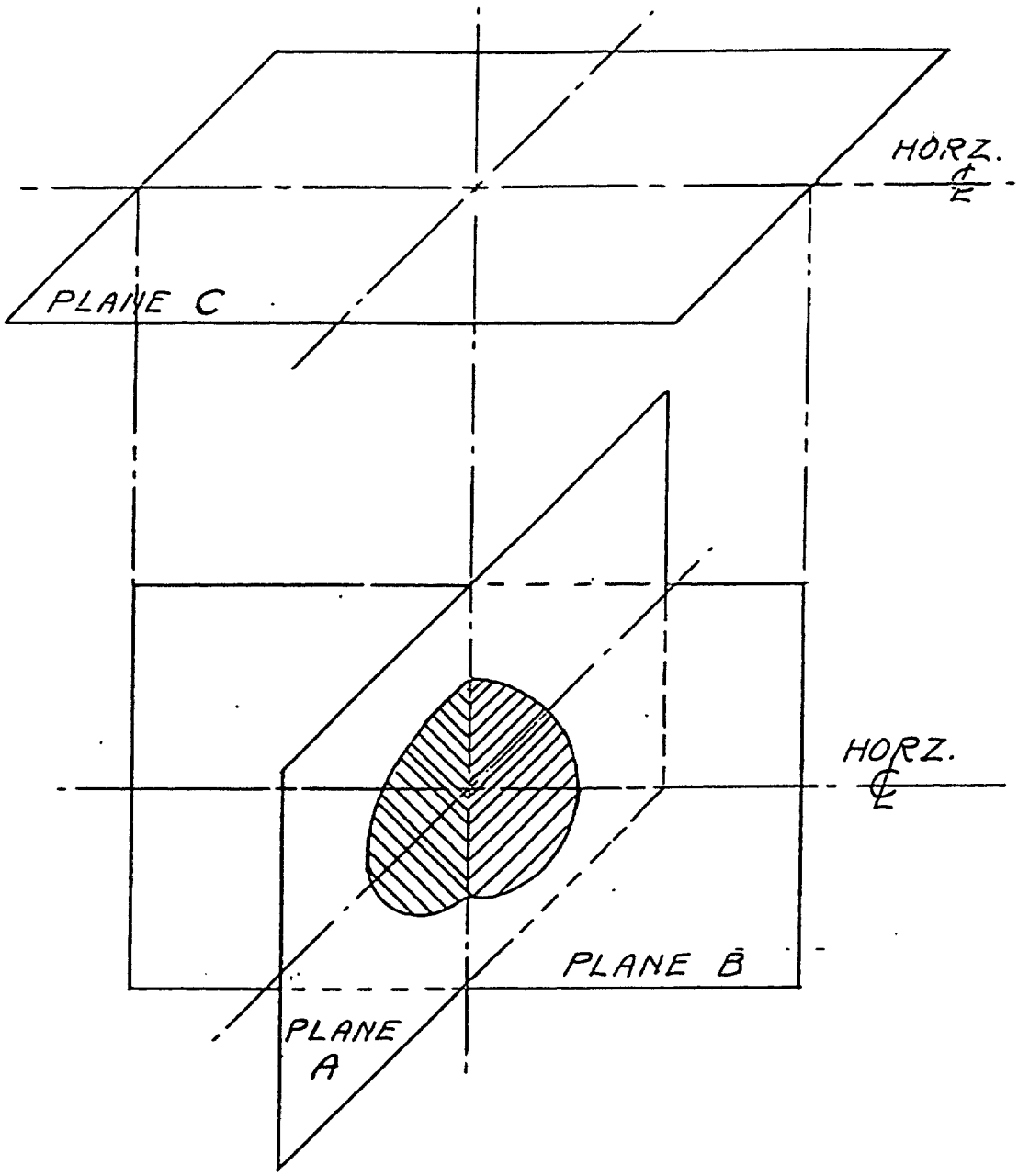


FIGURE 4. Planes

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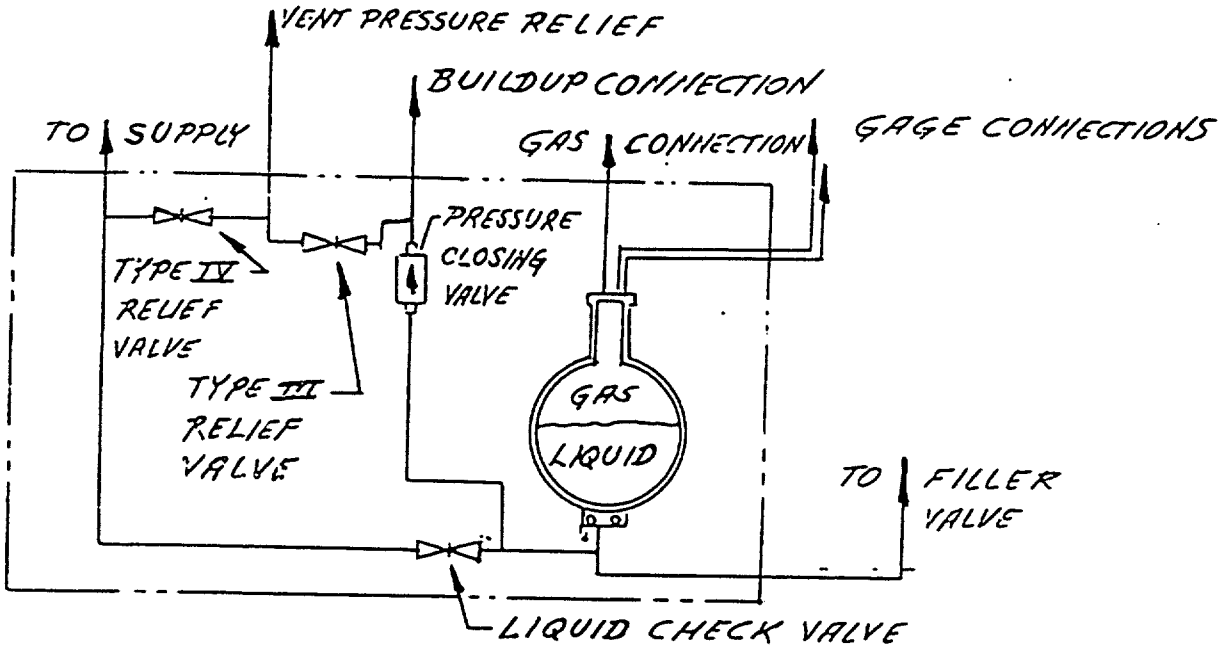
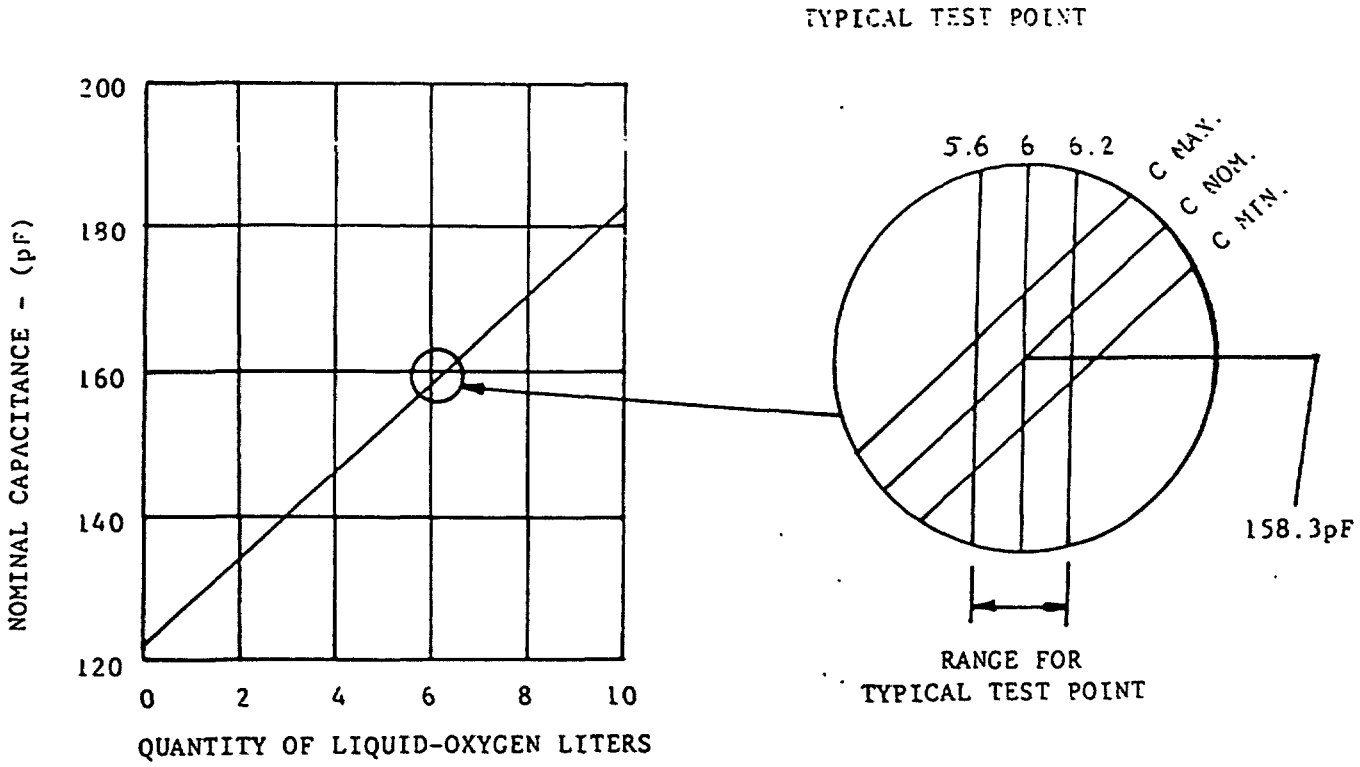


FIGURE 5. Schematic diagram

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Liquid oxygen (liters)	Nominal capacitance (pf)
0	123.5
2	135.1
4	146.7
6	158.3
8	169.9
10	181.5

FIGURE 6. Relationship between capacitance and volume of oxygen sensed

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	MIL-DTL-25663/3	950915

3. DOCUMENT TITLE: CONVERTER, LIQUID OXYGEN GCU-10/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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